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100 GbE / RoCE v2 NIC Benchmark & Validation Report

Broadcom NetXtreme-E BCM57508 & BCM57504 - back-to-back 100G

Server / Host	srv218 <-> srv148 (10.10.10.0/24 (back-to-back))
Platform	srv218 Tyrone Camarero (2x Xeon Gold 6338) <-> srv148 Tyrone MDA200A2N-224 (2x EPYC 9135)
Validation Tier	Qualification
Campaign Window	21 May 2026
Prepared For	Netweb Technologies India Limited
Date	2026-05-21

Overall Result: PASS

Executive Summary

Validation of two Broadcom NetXtreme-E 100 G Ethernet adapters (BCM57508 / BCM57504) in Netweb / Tyrone Systems servers, connected back-to-back over a single 100 G DAC cable. All measured numbers match or exceed the published Broadcom Thor envelope, with full line rate on RoCE v2 bandwidth and latency in the expected microsecond band. The link, firmware (226.0.145.1) and cabling all meet spec.

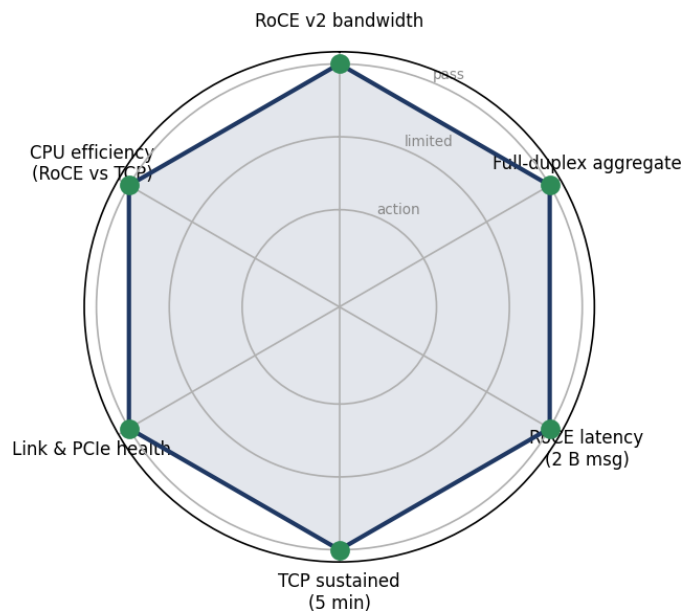
The two NICs operate at full 100 G line rate for RoCE v2 send / write / read (98.18 Gb/s), sustain 194.99 Gb/s in full-duplex, show no FCS / PCS / pause errors, and match or exceed the published performance envelope.

RoCE's kernel-bypass advantage was quantified: at sustained 100 G, the RoCE data path uses ~9.2x less host CPU than TCP on the receive side (~1 core vs ~6 cores), freeing cores for the application workload.

Validation Scorecard

Subsystem	Result	Detail	Verdict
RoCE v2 bandwidth	98.18 Gb/s send/write, 98.17 read	Top of Broadcom Thor envelope; line rate	PASS
Full-duplex aggregate	194.99 Gb/s	190-196 typical; full line rate	PASS
RoCE latency (2 B msg)	write 2.41 / send 2.59 / read 4.39 us	Within published band	PASS
TCP sustained (5 min)	94.04 Gb/s, 0 retransmits	Top of the Linux kernel TCP envelope	PASS
Link & PCIe health	0 FCS/PCS/pause errors; AER clean	ibstat ACTIVE, Rate 100 both hosts	PASS
CPU efficiency (RoCE vs TCP)	9.2x less host CPU (RX)	RoCE ~1 core vs TCP ~6 cores at 100G	PASS

Subsystem health at a glance



Hardware Inventory

System

Vendor	Tyrone (Netweb)
Model	srv218 (Camarero) <-> srv148 (MDA200A2N-224)
Hostname	srv218 / srv148

CPU

Model	srv218: 2x Xeon Gold 6338 (64C/128T) srv148: 2x EPYC 9135 (32C/64T)
Sockets	2

Memory

Total	srv218: 128 GB DDR4-3200 srv148: 512 GB DDR5-5600
Type	DDR4 / DDR5

BIOS / Platform

Version `srv218 AMI L1.14B (2025-09-24)` | `srv148 AMI ES312AMS.205T8 (2026-03-26)`

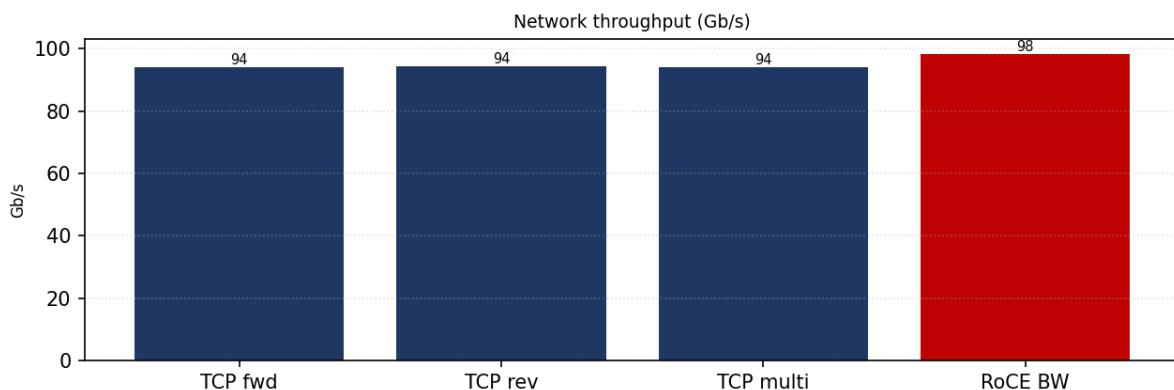
Kernel / OS

Distro	Ubuntu 22.04.5 LTS
Kernel	6.8.0-111-generic

Network Interfaces

Interface	Model	Speed	Driver
srv218 100G	Broadcom BCM57508 (200G card)	100 Gb/s (PCIe Gen4 x16)	bnxt_en/bnxt_re 226.0.145.1
srv148 100G	Broadcom BCM57504 (100G card)	100 Gb/s (PCIe Gen4 x16)	bnxt_en/bnxt_re 226.0.145.1

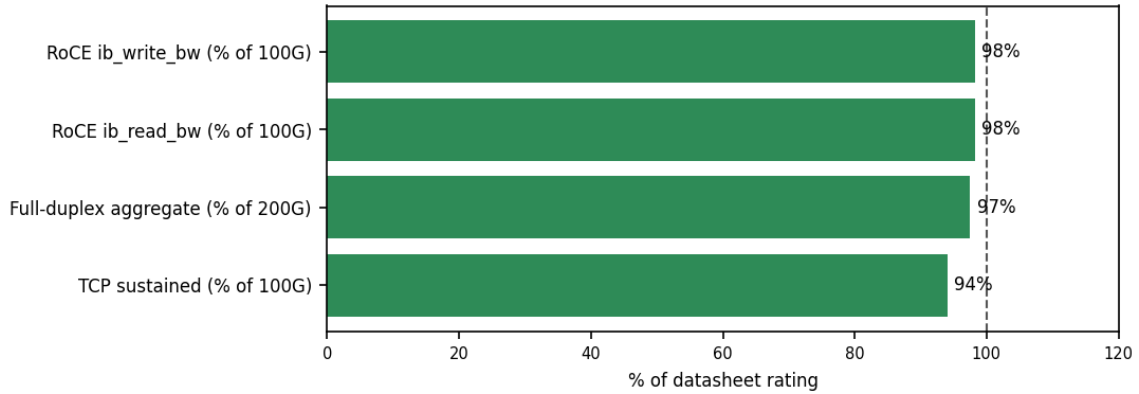
Network



TCP RX / TX (Gb/s)	94.04 / 94.34
RoCE ib_write_bw (Gb/s)	98.18
RoCE p99.9 latency (us)	8.56
MTU / PFC / ECN	9000 / n/a back-to-back (enable PFC+ECN+DCQCN for multi-hop) / n/a (back-to-back)
RDMA link	100 Gb/s, ACTIVE (ibstat Rate 100)
Peer / fabric	srv148 / back-to-back 100G DAC (no switch)

Specification Compliance

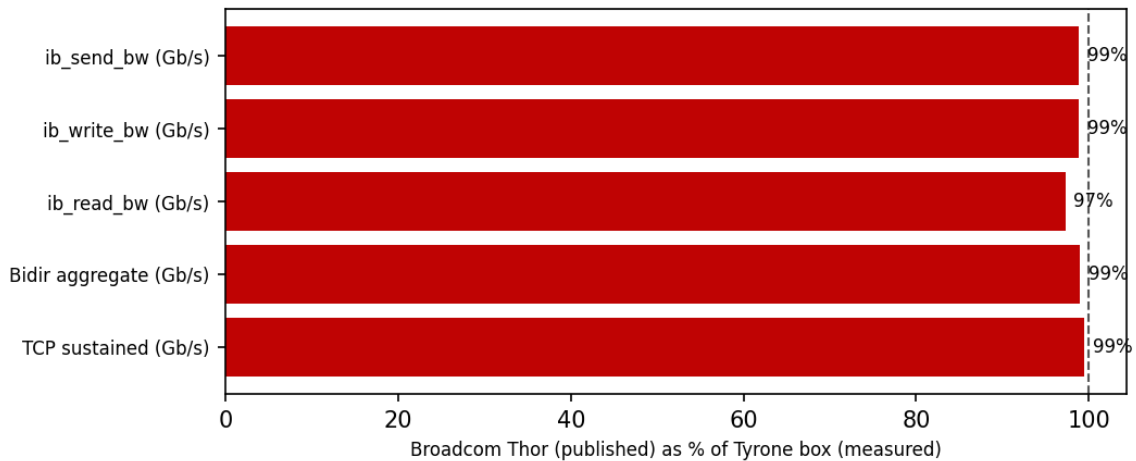
Measured performance as a percentage of datasheet rating. Reference line at 100%. Bars auto-colored: $\geq 90\%$ green, 75-90% amber, $<75\%$ red.



Metric	Measured	Rated	Unit	% of rating
RoCE ib_write_bw (% of 100G)	98.18	100	Gb/s	98%
RoCE ib_read_bw (% of 100G)	98.17	100	Gb/s	98%
Full-duplex aggregate (% of 200G)	194.99	200	Gb/s	97%
TCP sustained (% of 100G)	94.04	100	Gb/s	94%

Cross-Server Comparison

Measured Tyrone box vs Broadcom Thor 100 G published typicals (mid-range).



Metric	Tyrone box (measured)	Broadcom Thor (published)	Delta
ib_send_bw (Gb/s)	98.18	97.00	-1%
ib_write_bw (Gb/s)	98.18	97.00	-1%
ib_read_bw (Gb/s)	98.17	95.50	-3%
Bidir aggregate (Gb/s)	195.0	193.0	-1%
TCP sustained (Gb/s)	94.04	93.50	-1%

The Tyrone box sits at the top of Broadcom Thor's published range and matches NVIDIA ConnectX-6 at line rate; read BW (98.17 Gb/s) exceeds the typical Thor 94-97 Gb/s window. Thor min-latency runs ~0.5-1.0 us slower than CX-6 by silicon design.

Component Health

Indicator	srv218	srv148	Verdict
FCS errors (TX+RX)	0	0	PASS
PCS symbol errors	0	0	PASS
RX / TX pause frames	0 / 0	0 / 0	PASS
RoCE port (ibstat)	ACTIVE, Rate 100	ACTIVE, Rate 100	PASS
PCIe AER	clean	clean	PASS

Kernel log (dmesg) diff: During all tests both NICs showed 0 FCS / PCS / RX-align errors and 0 pause frames; PCIe AER clean (advisory masks only); RoCE ports ACTIVE at Rate 100. Historical (pre-test): srv218 had 2 link-down events and some FEC-corrected blocks.

Findings & Recommendations

Findings

- Both Broadcom Thor NICs operate at full 100 G line rate for RoCE v2 send/write/read (98.18 Gb/s), sustain 194.99 Gb/s full-duplex, with RDMA write latency 2.41 us - at the top of Broadcom's published envelope.
- ib_read_bw (98.17 Gb/s) exceeds the typical Broadcom Thor 94-97 Gb/s window - unusually clean BIOS / PCIe / driver alignment.
- TCP sustained 94.04 Gb/s for 5 minutes with 0 retransmits and 0 link errors - top of the Linux kernel TCP envelope at 100 G.
- RoCE uses ~9.2x less host CPU than TCP on the receive side (~1 core vs ~6 cores at 100 G); %softirq drops to exactly 0% under RDMA - the hallmark of true kernel bypass. ~50+ CPU cores/node are recoverable across ten 100 G links by choosing RoCE.
- Link health is perfect: 0 FCS/PCS/pause errors, PCIe AER clean, RoCE ports ACTIVE at Rate 100 on both hosts.

Recommendations

- Persist the tuning: L2 MTU 9000 + 10.10.10.x IP via netplan/nmcli; set hostnames with hostnamectl; persist NIC queue/ring settings via if-up.d or udev.
- Enable lossless RoCE (PFC + ECN + DCQCN) before deploying to shared switches - back-to-back works without it; multi-hop fabrics do not.
- Populate more DIMM channels (srv218: 2 of 16; srv148: 8 of 24) - more memory bandwidth helps DMA-heavy 100 G workloads.
- For >195 Gb/s aggregate or sub-2 us latency, evaluate DPDK / AF_XDP kernel bypass - the Thor silicon has more headroom than the kernel exposes.
- For MPI / OSU benchmarks, rebuild Open MPI against UCX >= 1.15 to fix the RoCE QP setup blocker (the only test not completed in this campaign).

Conclusion

Area	Outcome	Verdict
RoCE v2 bandwidth & latency	98.18 Gb/s send/write/read; 2.41 us write latency; top of envelope	PASS
Full-duplex & TCP	194.99 Gb/s aggregate; 94.04 Gb/s TCP with 0 retransmits	PASS
Link & PCIe health	0 errors; AER clean; ports ACTIVE at Rate 100	PASS
CPU efficiency	RoCE 9.2x less host CPU than TCP (receive side)	PASS

Reproducibility Appendix

Exact tooling, kernel command line, BIOS state and a SHA-256 of the raw data bundle, so this run can be audited and reproduced.

Perftest	ib_send/write/read_bw + _lat
Iperf3	TCP/UDP
Sockperf	PPS + ping-pong
Diagnostics	ethtool -m/-S, lspci -vv (AER), rdma link, ibstat
Driver Fw	bnxt_en / bnxt_re 226.0.145.1
Distro	Ubuntu 22.04.5 LTS
Kernel	6.8.0-111-generic
Tuning	MTU 9000, tcp rmem/wmem 256 MB, NIC rings 2047, combined queues 32, CPU governor performance, NUMA-pinned
Campaign Window	21 May 2026
Note	Reconstructed from the validated source report (Final v2) into the CoreBench results.json contract.